ECE343 Spring 2025

Objective 3.7: Transients



Lesson	Topics	Reading	Notes
1	Course Overview & Introduction	Ch.1	LO 1.1-1.2
Chapter 2: Electric and Magnetic Fields			
2	Fields, Parameters, Field Quantities	Ch. 2	LO 2.1-2.2
Chapter 3: Transmission Lines			
3	Lumped Element Model	3.1-3.4	LO 3.1
4	Telegrapher's Equations	3.5-3.6	LO 3.1
5	Characteristic Impedance & Wave Propagation	3.7-3.9	LO 3.2
6	Coaxial, Reflection Coefficient, VSWR	3.10,3.12-3.14	LO 3.2-3.3
7	Terminated Transmission Lines	3.15,3.17-3.19	LO 3.4
8	Impedance Matching	3.21-3.23	LO 3.4
9	Impedance Matching	3.21-3.23	LO 3.4
10	Smith Charts	Supplemental	LO 3.5
11	Smith Charts	Supplemental	LO 3.5
12	Smith Charts	Supplemental	LO 3.5
13	Smith Charts	Supplemental	LO 3.5
14	Transients	Supplemental	LO 3.6
15	Reflection/Transmission Coefficient Lab	Lab Packet	LO 3.7
Chapter 4: Vector Analysis			
16	Vector & Coordinate System Review	Ch. 4	LO 4.1-4.4 **GR due**
Chapter 5: Electrostatics			
17	Coulomb's Law	5.1-5.3	LO 5.1
18	Surface and Volume Charge	5.4	LO 5.1
19	Gauss' Law (Integral Form)	5.5-5.6	LO 5.2
20	Gauss' Law (Differential Form)	5.7-5.8	LO 5.2
21	Electric Potential	5.9-5.10, 5.15	LO 5.3

Ch 3 Learning Objectives

3.1 I can describe the origin of the transmission line equation, understand the method of developing a solution, and interpret the solution.

3.2 I can explain the physical meaning of and calculate the following quantities: propagation constant, phase constant, attenuation constant, characteristic impedance, VSWR, reflection coefficient, and transmission coefficient.

3.3 I can use the transmission line equation to calculate the voltage/current at an arbitrary point on a specified transmission line.

3.4 I can use the transmission line equation to match a load to a source using a length of transmission line.

3.5 I can identify the various quantities represented on the Smith Chart.

3.6 I can use a Smith Chart to design a matching network for a given source and load, using series components, parallel components, shorted stubs, and open stubs.

3.7 I can plot the transient response of a load connected to a source through a transmission line.

3.8 I can measure the reflection coefficient across a range of frequencies and interpret the measurement using a network analyzer.



TRANSIENTS

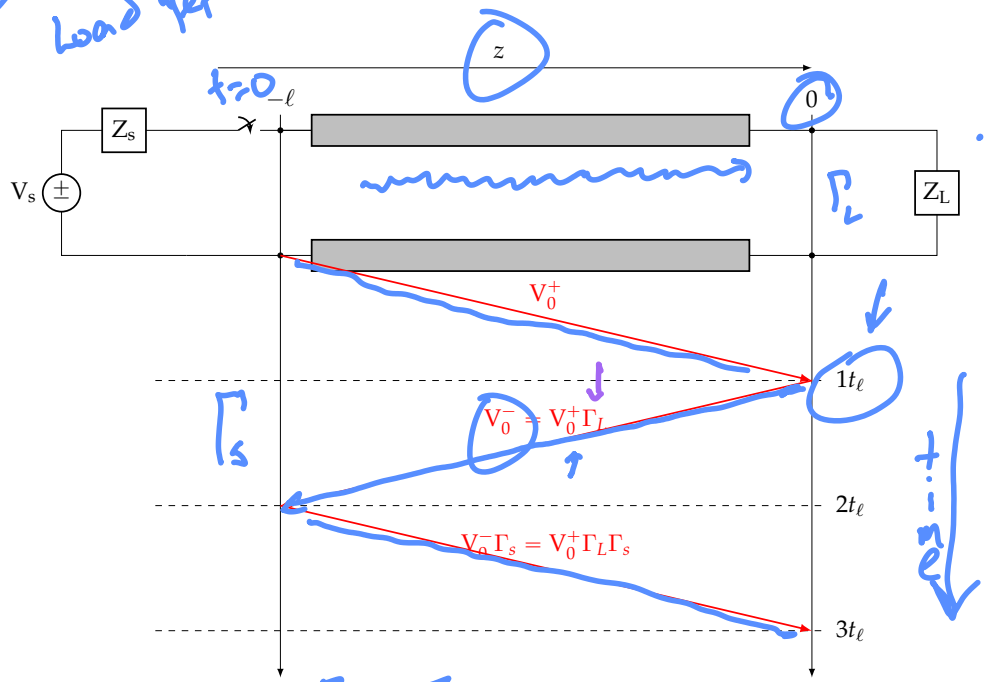
Method

- At $t = 0$, source voltage is divided between Z_0 and Z_0
- Wave propagates from $z = -\ell$ to $z = 0$ at velocity $v_p = c/\sqrt{\varepsilon_r} \approx 0.7c$
- Wave reflect at boundary of tx line & load (Γ_L)
- Wave propagates from load to source at velocity v_p
- Wave reflect at boundary of tx line & source Γ_s
- Voltage converges to steady of

$$V_L = V_s \left(\frac{Z_L}{Z_L + Z_s} \right)$$

215 0.Y.M.K

Doesn't "see" load yet



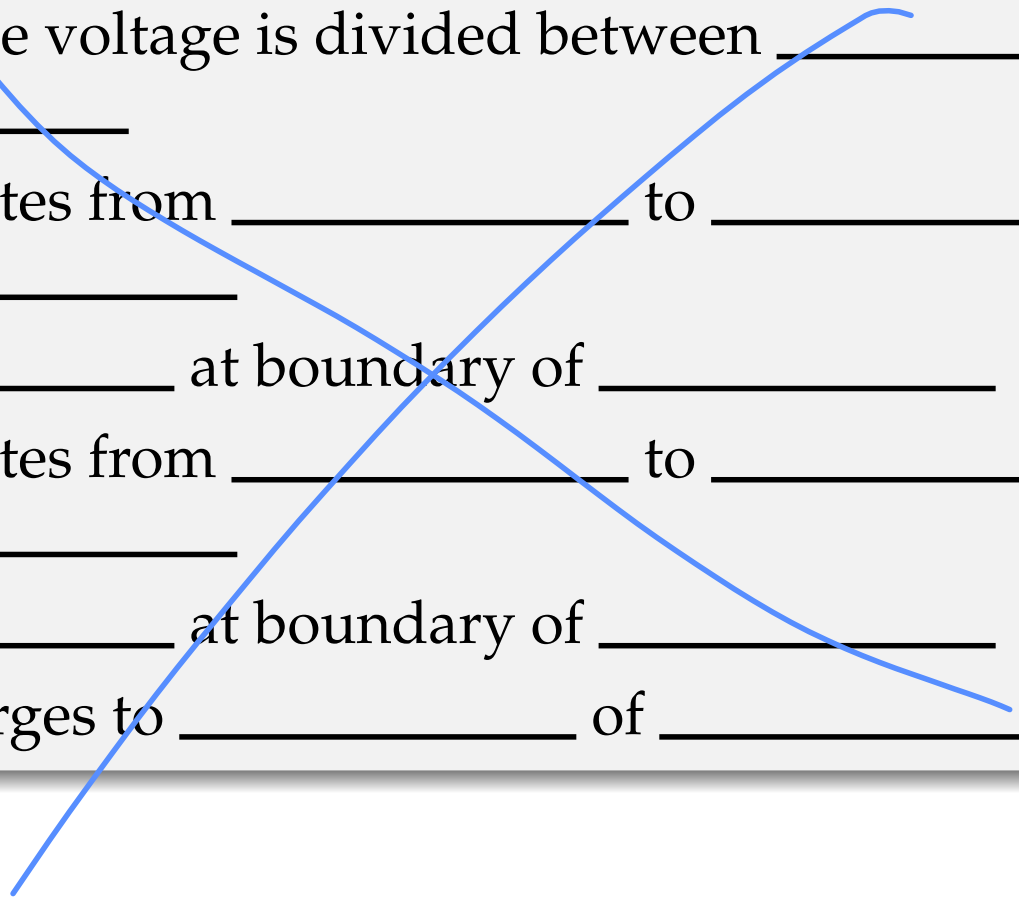
$$\Gamma_s = \frac{Z_0 - Z_0}{Z_s + Z_0}$$

$\Gamma_s, \Gamma_L < 1$

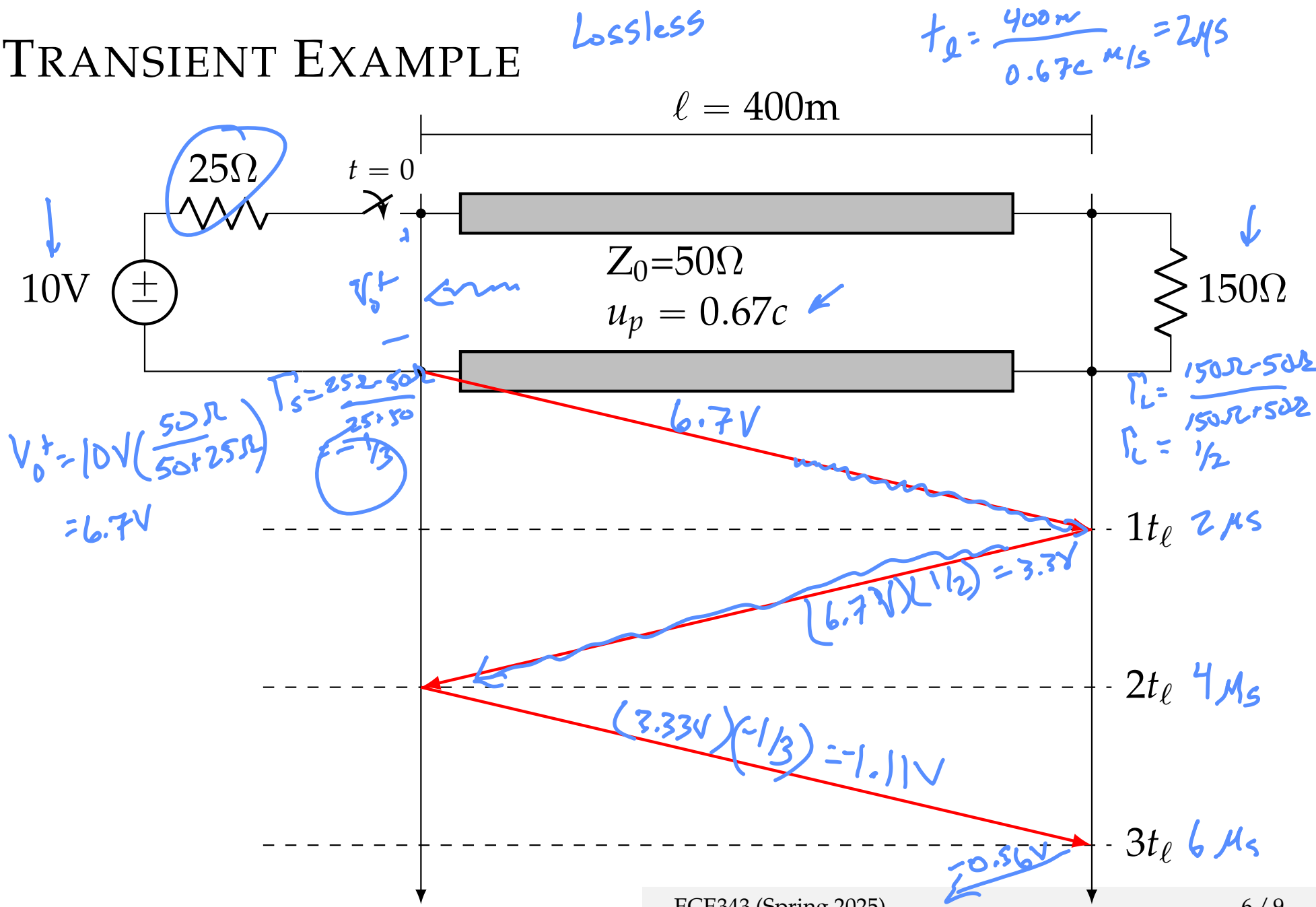
Transients

Method

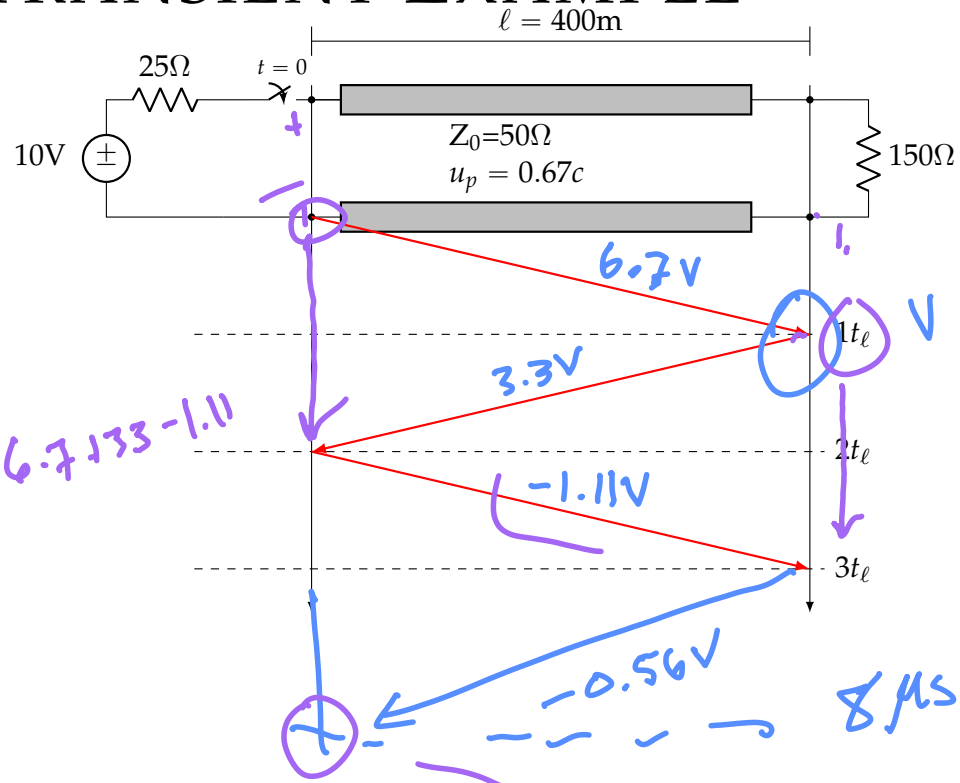
- At $t = 0$, source voltage is divided between _____ and _____
- Wave propagates from _____ to _____ at velocity _____
- Wave _____ at boundary of _____
- Wave propagates from _____ to _____ at velocity _____
- Wave _____ at boundary of _____
- Voltage converges to _____ of _____



Transient Example



Transient Example



6.7V

3.3V

6.7+3.3-1.11

-1.11V

-0.56V

8μs

10μs

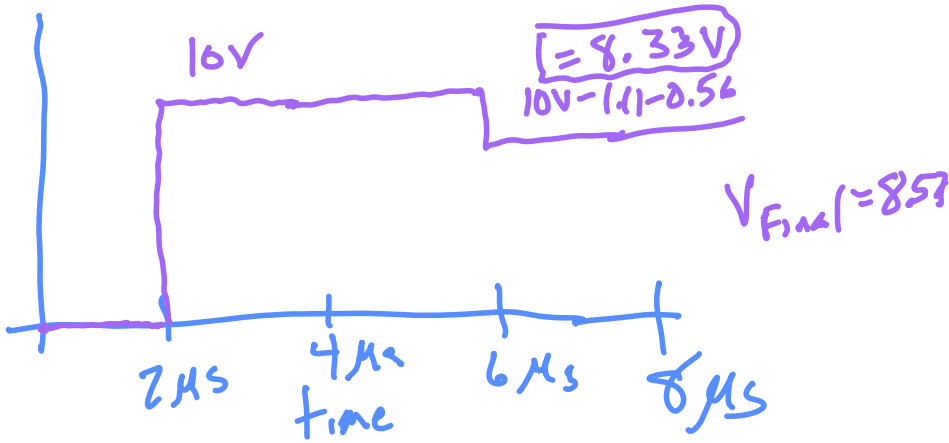
V

Source

$V_{Final} = 10 \frac{150}{175}$

$= 8.57$V

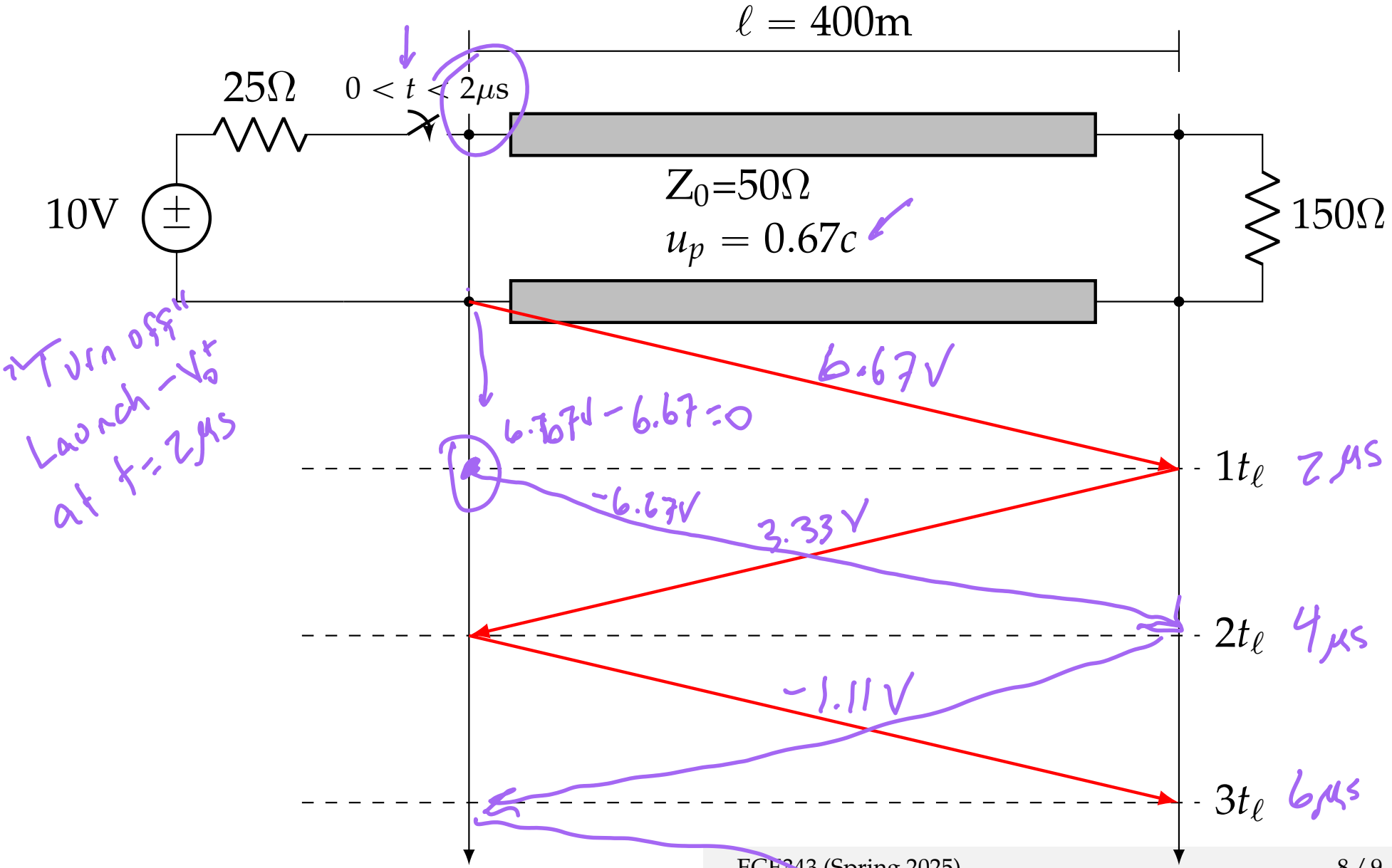
Load ($z = 0$):



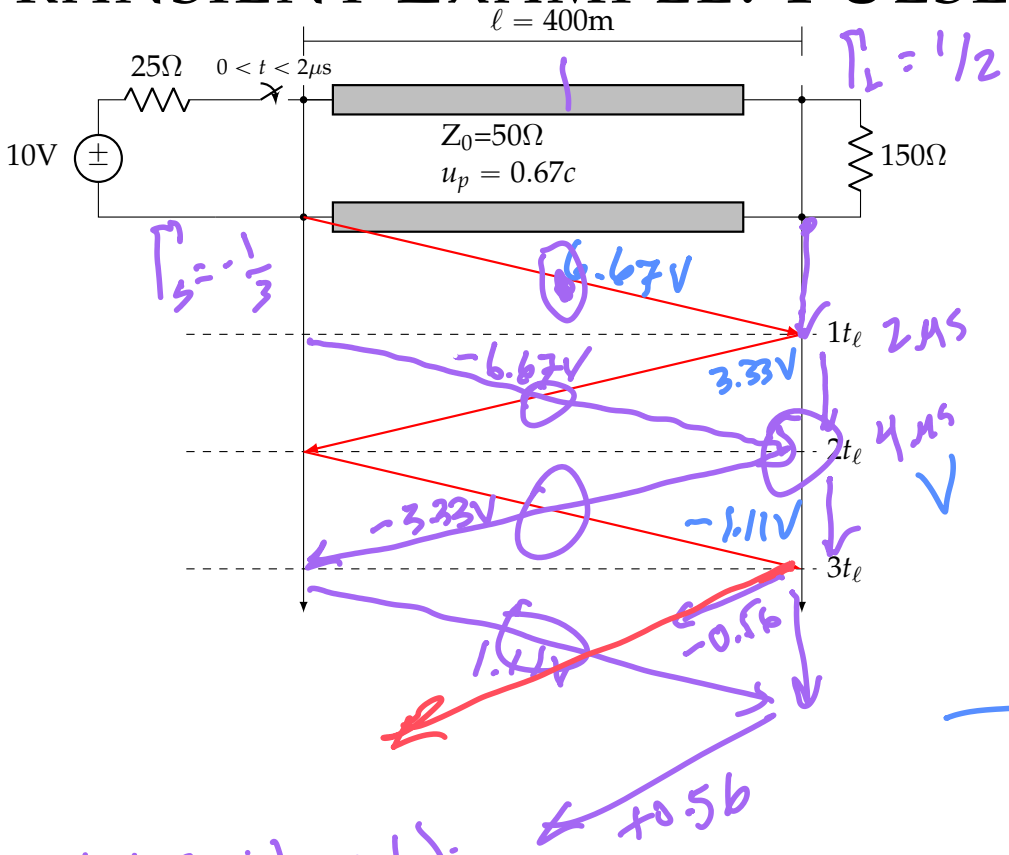
Source ($z = -\frac{\ell}{2}$)



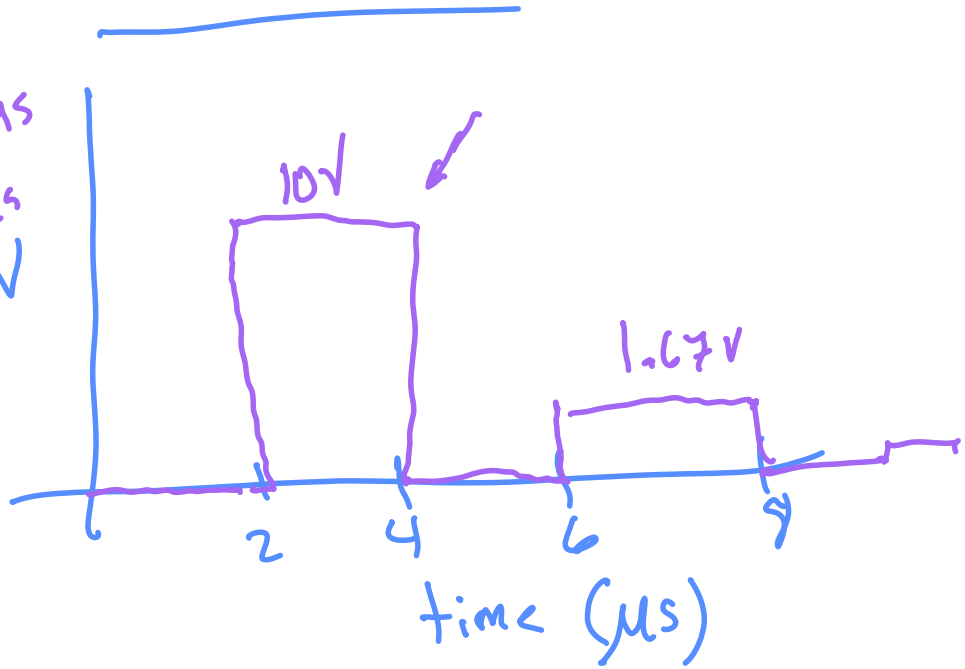
Transient Example: Pulsed Source



Transient Example: Pulsed Source



Plot (load):



Plot (midpoint):

